

Qihao Qian

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EDUCATION

University of California, San Diego (UCSD)

Master of Science in Electrical Computer Engineering, Machine Learning and Data Science

Sep 2024 – Dec 2026

San Diego, US

University of Electronic Science and Technology of China (UESTC)

Bachelor of Engineering in Software Engineering

Sep 2020 – June 2024

Chengdu, China

INDUSTRY EXPERIENCE

Tiktok (ByteDance)

MLE Intern

May 2026 – Aug 2026

Beijing, China

- Led post-training of **Qwen3-1.7B/4B/8B** for TikTok Tako query suggestion generation using training data constructed from millions of users' long- and short-term memories. Built the end-to-end training pipeline with **LLaMA-Factory**, **DeepSpeed ZeRO-3**, and **vLLM** (8×H100), resulting in **two production launches**.
- Built a preference optimization pipeline with **DPO**, reward-model filtering, trending-aware supervision, and diversity-aware data selection, improving **Inbox Query Recommendation PV CTR** by **22.71% (US)** and **12.75% (ROW)**.
- Designed an offline evaluation framework covering structured-output validity, semantic diversity, and standard LLM benchmarks (**MMLU**, **HellaSwag**, **IFEval**); demonstrated that **single-stage ORPO** outperformed the SFT→DPO pipeline while reducing **training compute** by **50%**, and further explored **On-Policy Distillation (OPD)**.
- Reduced input tokens by **51.5%** through semantic reformatting, solved 4B model format collapse via early stopping and small learning rate, trained a Draft Model for **643ms→552ms** real-time inference.

China Institute of People's Life Safety

AI Engineer Intern

Jan. 2023 – June 2023

Chengdu, China

- Synthesized **~5,000** instruction-style samples from PubMed abstracts using **GPT-3.5**; built a multi-stage cleaning pipeline (JSON validation, length/blacklist filtering, density checks, word-overlap removal) to improve SFT quality.
- Built a SentenceTransformer **semantic consistency filter** enforcing similarity between extracted fields and source sentences, reducing hallucinations and keeping Background/Objective/Methods/Results/Conclusion faithful to the text.
- Applied **paraphrase** augmentation and **JSON schema** perturbation to mitigate template overfitting, and **curriculum learning** (short-to-long abstracts) to improve training stability and generalization.
- Applied **LoRA** fine-tuning on LLaMA, with weighted loss on JSON-structural tokens and key medical fields (Results, Conclusion), ultimately lifting the JSON validity rate to **95%+** and improving field-level Token-F1 by **20%**.

DJI

MLE Intern

June 2022 – Dec 2022

Shenzhen, China

- Improve the **30-minute per scene** bottleneck in autonomous-driving lane annotation to **under 5 minutes** by building an AI-assisted platform where users annotate lanes via rough bounding boxes.
- Trained **PointNet++** on Toronto3D, and introduced **Random Drop**, **Global Scaling & Rotation**, and lane **Oversampling** data augmentation to handle sparse real-world lane point clouds.
- Achieved lane segmentation **F1 85.3%** on the test set; built a GPU inference service based on **PyTorch** with custom **CUDA extension** and deployed with **Docker**, reliably supporting large-scale internal batch pre-annotation tasks.

RESEARCH EXPERIENCE

Online Euclidean SDF & Vision-Language Mapping

UCSD Existential Robotics Lab | IROS 2026 (First Author) | AutonomousRobots (Third Author)

Jan 2025 – Mar 2026

San Diego, US

- Led the proposal and implementation of **OREN**, a hybrid data structure(Octree Prior + Neural Residual) that balances online mapping efficiency and geometric accuracy, producing a continuous high-precision distance field.
- Designed a **semi-sparse octree** with gradient-enhanced interpolation; compared with a sparse octree and trilinear interpolation, reduced **SDF MAE** by **45%** and gradient MAE by **38%**.
- Proposed a **projection loss** to supervise free space, improving Chamfer Distance by **6×** and F1 score by **14%**.
- Built **OREN-VL**: extracted dense **semantic features** from RGB using **TIPS-v2**, fused them online into the octree via depth back-projection and trilinear interpolation, enabled open-vocabulary 3D semantic queries on the surface.
- Built the training, evaluation, and **ROS 2** online mapping system; validated on synthetic/real-world datasets and **quadrotor UAV**; combined with Distance-Accelerated Motion Planning for real-time distance queries and path planning.